

Mallika Laasya Ramachandrani

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EDUCATION

University of Wisconsin – Madison
B.S. Computer Science & Economics

Madison, WI
Expected May 2028

- Relevant Coursework: Programming 3, Linear Algebra, Statistics, Matrix Methods in Machine Learning

TECHNICAL SKILLS

Languages & Frameworks: Java, Python, TypeScript, JavaScript, SQL, HTML/CSS, React, Next.js, FastAPI, Node.js, Tailwind CSS, scikit-learn

Cloud, Backend & Dev Tools: Supabase, Firebase, PostgreSQL, SQLite, REST APIs, Docker, Git, GitHub Actions, CI/CD, Postman

AI/ML: Claude API, Groq API, OpenAI Whisper, scikit-learn, LLM Integration, Prompt Engineering

EXPERIENCE

Undergraduate Learning Center – Incoming CS Tutor
University of Wisconsin – Madison

Aug. 2026 – Present
Madison, WI

- **Selected to tutor CS 200/300** covering core Java fundamentals, OOP inheritance, recursion, ArrayLists, HashMaps, and Big-O time complexity analysis
- Conduct weekly group office hours and one-on-one sessions, translating complex algorithmic concepts into concrete, debuggable examples for students from varied technical backgrounds
- Developing targeted supplemental problem sets addressing common misconceptions in data structures and OOP to support student comprehension beyond lecture material

Pain Context Innovation Lab – Research Assistant
University of Wisconsin – Madison | PI: Dr. Shinye Kim

Jan. 2026 – Present
Madison, WI

- **Reduced average time-to-clinic-match by ~40%** by architecting and deploying the clinic navigation module for a patient-facing web platform serving 100s of chronic pain patients
- **Cut average user lookup time from 5+ min to under 60 sec** by building responsive, real-time search and filter UI components in **React** with dynamic state management and debounced API calls
- **Integrated the NPPES NPI REST API** to dynamically query and filter **1M+ clinician records** by insurance type, specialty, and geographic location, surfacing the most relevant providers for each patient
- Collaborated with a 5-person cross-functional team of clinicians and developers across bi-weekly sprints, managing feature branches and conducting pull request reviews on GitHub

PROJECTS

Vocalis | Next.js, TypeScript, FastAPI, Groq, Whisper, Docker, Firebase | [GitHub](#)

June 2026

- **Achieved <2s transcription latency** on responses up to 3 minutes by engineering an end-to-end voice pipeline integrating **OpenAI Whisper** with the Web Audio API for real-time in-browser audio capture and streaming transcription
- **Delivered structured feedback in under 3 seconds** by designing a **scikit-learn** scoring model evaluating interview answers across 4 dimensions: clarity, relevance, depth, and communication quality
- Built a full-stack AI mock interview platform for 100s of students, generating role-specific technical and behavioral questions across 3 engineering disciplines via **Groq-hosted LLMs** and a RESTful FastAPI backend
- Containerized the full backend (FastAPI + SQLAlchemy + SQLite) with **Docker Compose**; deployed with a Firebase-authenticated CI/CD pipeline enabling reproducible one-command local setup and consistent staging environments

VitalLens | React, TypeScript, Supabase, Claude API, Recharts | [GitHub](#)

April 2026

- **Won 2nd Place at the BadgerAI Hackathon** for VitalLens, a full-stack, patient-facing health platform built end-to-end during the event
- **Cut patient appointment prep time from 30+ min to under 5 min** by automating personalized doctor prep question generation derived from longitudinal biomarker trends visualized with Recharts across multiple visits
- Engineered a PDF parsing pipeline that extracts and classifies 20+ biomarker types into structured JSON output via **Claude Sonnet 4.6**, serving 100s of users with actionable, plain-English health insights
- **Ensured zero cross-user data exposure** by implementing Supabase Auth with Row-Level Security policies and PostgreSQL Edge Functions, achieving HIPAA-aligned per-user data isolation

CAMPUS INVOLVEMENT

WACM – Undergraduate Activities Chair
University of Wisconsin – Madison

Jan. 2026 – Present
Madison, WI

- Selected as Undergraduate Activities Chair for Fall 2026; curate a monthly technical speaker series for 50+ members featuring engineers from Google, Microsoft, and Amazon

Women in Business Technology – VP of Systems
University of Wisconsin – Madison

Jan. 2026 – Present
Madison, WI

- Elected VP of Systems for Fall 2026; architect and maintain club website for a 100+ member organization, shipping 10+ feature updates per semester and integrating professional networking resources
- Connected 100+ members with 20+ industry professionals through platform-driven outreach and coordinated networking events each semester